COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010-DATA STRUCTURE AND ALGORITHMS	**SECTION:**	CPE21S4
DATE PERFORMED:	7/17/26	**DATE SUBMITTED:**	7/17/26
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ACTIVITY NO. 2.1 **Arrays, Pointers and Dynamic Memory Allocation**

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: Implement static and dynamic memory allocation & ILO B: Create dynamically allocated objects using pointers and arrays In this activity, we will explore static and dynamic memory allocation through by utilizing arrays and other components mentioned beforehand, this will be for the implementation of a student list with the students' names and age. To begin, we will include pertinent libraries for input/output stream and strings (which we will be using in this activity).

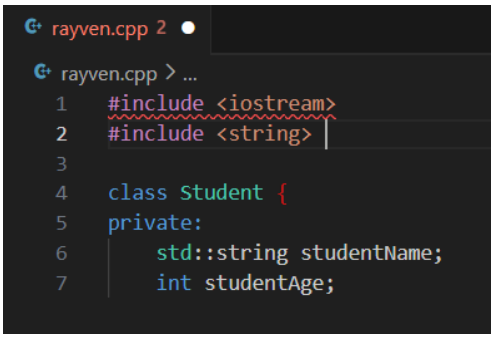
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

5. Procedure

In this activity, we will explore static and dynamic memory allocation through by utilizing arrays and other components mentioned beforehand, this will be for the implementation of a student list with the students' names and age. To begin, we will include pertinent libraries for input/output stream and strings (which we will be using in this activity). #include #include

Now we will create the student class with private attributes studentName and studentAge.



Then, as discussed beforehand, we have to define the constructor and the big three: the destructor, copy constructor and copy assignment operator. We will assign default values for the parameters of our constructor in the event that it is uninitialized.

```
public:
    Student(std::string newName = "John Doe", int newAge = 18) {
        studentName = std::move(newName);
        studentAge = newAge;
        std::cout << "Constructor Called." << std::endl;
    }
```

The big three is then defined. We will add an output stream for each to observe when each of the following functions are implicitly or explicitly called.

```
Student(std::string newName = "John Doe", int newAge = 18) {
    studentName = std::move(newName);
    studentAge = newAge;
    std::cout << "Constructor Called." << std::endl;
}

~Student() {
    std::cout << "Destructor Called." << std::endl;
}

Student(const Student &copyStudent) {
    std::cout << "Copy Constructor Called" << std::endl;
    studentName = copyStudent.studentName;
    studentAge = copyStudent.studentAge;
}

void printDetails() {
    std::cout << this->studentName << " " << this->studentAge << std::
```

The driver program is now to be defined. We want to utilize the main function to simply show the static and dynamic allocation. However, we will explore the initial goal of creating instances of our Student class.

```
int main() {
    const size_t j = 5;

    Student studentList[j] = {};
    std::string namesList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
    int ageList[j] = {15, 16, 18, 19, 16};

    for (int i = 0; i < j; i++) {
        Student *ptr = new Student(namesList[i], ageList[i]);
        studentList[i] = *ptr;


    for (int i = 0; i < j; i++) {
        studentList[i].printDetails();
    }

    return 0;
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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